Landsat and Sentinel-2 based floods maps of the Central European Flood event in September 2024

Proposal: Interdisciplinary Project in Data Science

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June 2025

1 Abstract

This project aims to develop a machine learning-based approach for assessing flooded areas using satellite imagery, with a focus on the Central European flood event of September 2024, primarily caused by Storm Boris. The event led to significant flooding across Austria, Poland, Slovakia, and Germany, resulting in strong economic impacts. Leveraging multispectral optical imagery from the Landsat and Sentinel-2 satellite series—including RGB and near-infrared (NIR) bands — the project focuses on estimating the likelihood of water presence at the pixel level, extended by more in-depth analysis of the model performance in flooded areas.

The core of the methodology involves training a convolutional neural network for semantic segmentation (DeepLabV3+) to detect water-covered areas, extended by Monte Carlo dropout to generate uncertainty-aware predictions. This probabilistic framework enables the production of confidence maps that quantify the reliability of flood detection across spatial regions. The workflow includes data collection, preprocessing (e.g., cloud masking, tiles splitting), and feature engineering (e.g., NDWI), followed by model training and evaluation using standard metrics such as IoU, F1 score, Precision, Recall, and Dice Coefficient.

Ultimately, the project aims to assess the utility of optical satellite data in flood mapping and demonstrate the model's ability to distinguish floodwaters from other land surfaces with quantifiable confidence.

2 Motivation and Problem Statement

In September 2024, Central and Eastern Europe experienced one of the most devastating natural disasters in recent years when Storm Boris triggered a strong rainfall—up to 500 mm in under 48 hours—resulting in widespread flooding. The event affected countries including Austria, Czech Republic, Slovakia, Poland, Romania, Hungary, and Germany. With 27 confirmed fatalities and economic losses exceeding several billion euros, the disaster highlights the importance of effective flood monitoring tools.

Satellite-based Earth observation is a powerful tool for disaster monitoring, but accurately identifying floodwaters remains challenging. Radar sensors like Sentinel-1 can operate under cloud cover, but optical sensors (e.g., Landsat and Sentinel-2) offer better spatial and spectral resolution and are more readily available. However, their effectiveness is limited by atmospheric conditions such as cloud cover, making water classification more complex.

Traditional flood detection methods often rely on binary classification (flooded vs. non-flooded), providing no indication of prediction confidence. This restricts the usefulness of such tools in real-world emergency scenarios where uncertainty matters. This project is motivated by the need to go beyond deterministic outputs by developing a machine learning model that estimates water likelihood at the pixel level, integrating multispectral satellite data and deep learning-based segmentation with an uncertainty estimation technique - Monte Carlo dropout.

3 Research Questions

1. **To what extent can high-quality waterbody masks be generated using NDWI and Sentinel-2 Scene Classification Layer (SCL) data?**

   Initial waterbody labels will be generated using thresholding on the NDWI index. The resulting masks will be evaluated through manual visual inspection (Sentinel-based part of the dataset) and quantitative comparison with reference datasets, e.g., JRC Global Surface Water (Landsat-based part of the dataset). The primary evaluation metric will be the Correction Rate, defined as the proportion of masks that required manual refinement after automatic generation. This assessment aims to determine whether the semi-automated labeling approach produces annotations of sufficient quality for model training.

2. **How accurately can the proposed deep learning model identify water-covered areas in satellite imagery?**

   This question will be addressed by evaluating the final trained model on the labeled test dataset. The model's performance in detecting water-covered areas will be measured using standard semantic segmentation metrics, including accuracy, precision, recall, F1 score, and Intersection over Union (IoU). These metrics will quantify the model's effectiveness in identifying waterbodies and flooded regions.

3. **To what extent can a model trained on Sentinel-2 data be transferred to Landsat imagery?**

   To assess cross-sensor generalization, the model trained on Sentinel-2 data will be directly applied to Landsat images. Performance will be evaluated using the same segmentation metrics (IoU, F1 score, precision, recall), and results will be compared against those obtained on Sentinel-2 test data. A significant drop in performance will indicate limited transferability.

4. **Can uncertainty estimates be used to improve the reliability of flood classification?**

   The analysis will focus specifically on images captured during flood events. Uncertainty will be estimated using Monte Carlo Dropout. At multiple uncertainty thresholds, low-confidence pixels will be filtered out, and segmentation quality will be re-evaluated on the remaining, high-confidence areas. Metrics such as precision, recall, F1 score, and Intersection over Union (IoU) will be computed for multiple thresholds.

4 Methodology

This project follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology to ensure a structured, iterative, and reproducible approach to flood mapping using satellite imagery. The main phases include domain understanding, data preparation, modeling, evaluation, and deployment of results in a format suitable for real-world interpretation.

4.1 Business Understanding

The project began with a review of the September 2024 Central European flood event, analyzing its scope and affected regions. In addition, relevant studies [2], [3] were reviewed to understand how similar research is structured and conducted.

4.2 Data Understanding

The next step involved a detailed assessment of the data sources required for effective flood detection. Multispectral imagery from Sentinel-2 and Landsat-8/9 was selected due to its high spatial resolution, frequent revisit rates, and broad geographic coverage. Several challenges were identified early on, including the presence of cloud cover in optical imagery, spectral confusion between water, snow, and urban surfaces, and the need for accurate, event-specific ground truth labels to support supervised learning.

4.3 Data Preparation

The dataset creation process follows the DiRS principles for remote sensing benchmark datasets—ensuring diversity, richness, and scalability. It includes the following steps:

- Data Collection: Half the dataset will consist of Sentinel-2 imagery, and the other half from Landsat (surface reflectance product). Cloud-covered or low-quality scenes are filtered using the QA bands and the Scene Classification Layer (SCL).
- Initial Mask Generation: Water and cloud masks are generated using spectral indices such as NDWI and NDSI, with pixel-level thresholds. For Landsat, where misclassification with ice/snow may occur, manual corrections will be applied. For Sentinel, to avoid ice/snow misclassification the Scene Classification Layer (SCL) will be used.
- Label Refinement: Each image will be manually inspected (with the help of `napari` Python library), and masks corrected where necessary. Persistent waterbodies will be augmented using data from known sources (e.g., Global Surface Water dataset).
- Harmonization: Landsat images will be resampled to 10m resolution using the GDAL library to match Sentinel-2 data, ensuring consistent input for the modeling pipeline.

Reproducibility will be ensured by providing full documentation and metadata for the dataset, including preprocessing steps, annotation guidelines, and versioning.

4.4 Modeling

The modeling phase will proceed in the following stages:

- Step 1: A DeepLabV3+ model will be trained to perform binary segmentation of water and non-water areas.
- Step 2: The trained model will be improved by optimizing the parameters and architecture.
- Step 3: To capture predictive uncertainty, the DeepLabV3+ will be extended using Monte Carlo Dropout (MC Dropout). During inference, multiple forward passes will be performed to produce per-pixel mean probabilities and standard deviations, enabling the creation of confidence maps.

Hyperparameters will be tuned using validation subsets, and stratified spatial splits will be used to ensure generalizability across different regions.

4.5 Evaluation

Model performance will be evaluated using accuracy, Precision, Recall, F1 Score, and Intersection over Union (IoU). To test generalization, a model trained on Sentinel-2 will be evaluated on Landsat images, measuring performance drops and adaptability across sensors. Additionally, the ability of the model to detect flood water will be inspected.

4.6 Deployment

While no live deployment is planned for this project, all results will be delivered in the form of a comprehensive final report and a publicly accessible GitHub repository containing the source code, documentation, and dataset preparation pipeline. More details are provided in the *Expected Results* section.

5 Expected Results

At the conclusion of the project, the following results are anticipated:

- A **curated and labeled dataset** of Sentinel-2 and Landsat imagery, annotated with high-quality masks (cloud, water, etc.). The dataset will be created according to the DiRS principles.
- A **trained deep learning model** capable of performing semantic segmentation of water bodies, including flooded areas. The model will output both binary classification masks and probability maps for the water class, providing a confidence score for each pixel.
- **Visualization outputs**, including side-by-side comparisons of input images, annotated masks, model predictions, and confidence maps for images captured shortly after the Boris flood event.
- A **report** providing a comprehensive description of all steps taken during the project and the results obtained will be submitted. In addition, a **GitHub repository** containing all relevant code will be made available.

6 Related Domain-specific Lectures

120.110 Data Retrieval in Earth Observation. The course has already been completed.

References

[1] Dan López-Puigdollers, Gonzalo Mateo-García, Luis Gómez-Chova. *Benchmarking Deep Learning Models for Cloud Detection in Landsat-8 and Sentinel-2 Images.* March 2021. In: Remote Sensing 13.5. ISSN: 2072-4292. (DOI: 10.3390/rs13050992.).

[2] Wieland, M., & Martinis, S. *A modular processing chain for automated flood monitoring from multi-spectral satellite data.* October 2019. In: Remote Sensing, 11.19. ISSN: 2072-4292. (DOI: 10.3390/rs11192330.).

[3] Wieland, M., Martinis, S., Kiefl, R., & Gstaiger, V. *Semantic segmentation of water bodies in very high-resolution satellite and aerial images.* March 2023. In: Remote Sensing of Environment, 286, 113402. ISSN: 0034-4257. (DOI:10.1016/j.rse.2023.113402.).

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[6] Copernicus Data Space Browser. *Explore and Download Copernicus Sentinel Data.* Accessed June 2025. https://dataspace.copernicus.eu/